



FOUR-COURSE SYSTEM

## COURSE 3 - DATA ENGINEERING 2026

# C3-044B ClinicQueue Orchestration Gate

SEALED fresh full retest - explicit assignment only.

|              |                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------|
| Artifact     | C3-044B                                                                                       |
| Status       | 2026 - LEARNER EDITION                                                                        |
| Phase        | C3-03 Orchestration & Reliable Workflows   13 lessons   34-46 focused hours                   |
| Prerequisite | C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here. |
| Brand        | Charcoal #1F2937   Teal #14B8A6   Soft Blue #3B82F6   AMOLED #000                             |

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.

# ClinicQueue scenario

ClinicQueue needs a reliable daily visit-event workflow under a separate domain and data shape. This is a fresh full retest, not a changed copy of Gate A.

**Gate B/C is not available merely because a prior Gate failed.**

## Independent rules

- Use supplied case ZIP; do not edit fixtures to make controls pass.
- External lesson/review/video answers are closed during attempt.
- State logical interval and side-effect contract before implementation.
- Save actual DAG/code/config plus run/task/interval and independent reconciliation evidence.
- No secret/credential value belongs in code or evidence.

## Required deliverables

- Fresh DAG with explicit wait -> extract -> validate -> load/merge -> reconcile -> publish boundaries.
- Correct daily logical interval + timezone-aware start; prove historical rerun path stability.
- Separate Params/config/Connection IDs/secrets; sanitize evidence.
- One bounded transient retry + one deterministic failure that stays failed.
- Bounded readiness wait + missing/late input recovery.
- Named pool for case sink + deliberately bounded backfill concurrency.
- Process/reprocess all supplied dates; prove stable version semantics and same-date idempotence.
- Layered predeploy tests/static checks + real learner-Airflow discovery/run evidence.
- Diagnose one injected incident from exact run/task/interval evidence; perform smallest safe recovery.
- Concise operator runbook + post-recovery reconciliation.

## Competency Gate - every required competency must pass

| Competency                                                        | PASS evidence                                                         |
|-------------------------------------------------------------------|-----------------------------------------------------------------------|
| DAG architecture and meaningful task boundaries                   | Observed, validated, reproducible and explainable on this fresh case. |
| Interval correctness and historical partition mapping             | Observed, validated, reproducible and explainable on this fresh case. |
| Failure classification, bounded retry/timeout                     | Observed, validated, reproducible and explainable on this fresh case. |
| Idempotent same-date rerun and stale-safe historical reprocessing | Observed, validated, reproducible and explainable on this fresh case. |
| Readiness wait and scarce-resource controls                       | Observed, validated, reproducible and explainable on this fresh case. |
| Testing/discovery/troubleshooting evidence                        | Observed, validated, reproducible and explainable on this fresh case. |
| Operator runbook and professional handoff                         | Observed, validated, reproducible and explainable on this fresh case. |

**PASS only if every required competency passes AND zero critical failures remain. Every required competency stands on its own; one failure blocks PASS.**

## Evidence packet

| Evidence                               | Your saved proof/path/note |
|----------------------------------------|----------------------------|
| DAG/task graph                         |                            |
| Schedule + interval mapping            |                            |
| Sanitized config/connection/pool       |                            |
| Normal run evidence                    |                            |
| Transient retry evidence               |                            |
| Missing/late input behavior            |                            |
| Historical backfill/reprocess evidence |                            |
| Same-date/partial rerun idempotence    |                            |
| Pre-deploy test evidence               |                            |
| Reconciliation controls                |                            |
| Incident troubleshooting note          |                            |
| Runbook/handoff                        |                            |